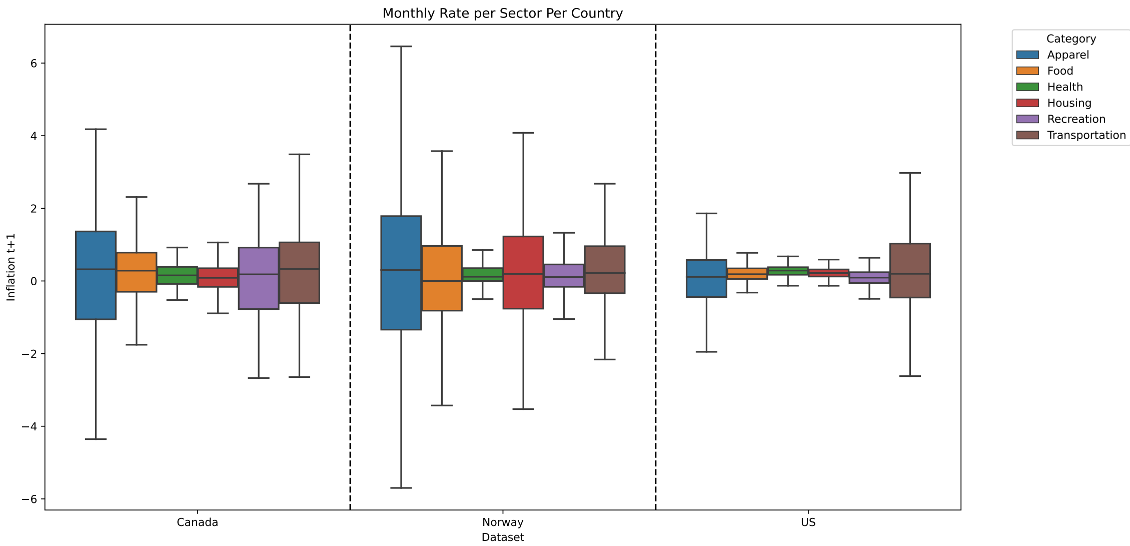
Figure 4.2 presents a box plot showing the distribution of CPI change rates across various sectors. It is evident that certain sectors, like Apparel and Transportation, exhibit greater volatility than others across all markets. This higher volatility is likely to make predictions for these sectors more challenging, as anticipated.

Figure 4.2. ***Monthly Rate per Sector Per Country***



4.4.1.1 Data Augmentation

As part of the pre-processing process we addressed the issue of missing weights in the Canadian dataset by employing a linear regression-based imputation method. In this approach, the prices of child categories were used as the independent variables (predictors), while the price of the parent category was treated as the dependent variable (target). This regression model allowed us to estimate the missing weights by leveraging the relationship between the prices of child and parent categories, thereby ensuring a consistent and reliable imputation strategy. Since the model's loss function incorporates the weights of each child category, this imputation step was a critical part of the preprocessing pipeline.

5 Evaluation and Results

We evaluate the performance of the BiHRNN and compare it to well-established baselines for inflation prediction, as well as some additional machine learning approaches. We use the following notation: Let x_t be the CPI log-change rate at month t . Models for $\hat{x}_t$ are considered as an estimate for x_t based on historical values. Furthermore, we denote the estimation error at time t by ε_t . Hyper-parameters were set using Bayesian-inspired optimization procedures.

5.1 *Baseline Models*

We compare Bidirectional HRNN with the following CPI prediction baselines:

1. **Autoregression (AR) -** The AR(ρ) model estimates $\hat{x}_t$ based on the previous ρ timestamps using the following equation: $\hat{x}_t = \alpha_0 + \left(\sum_{i=1}^{\rho} \alpha_i x_{t-i}\right) + \varepsilon_t$, where $\{\alpha_i\}_{i=0}^{\rho}$ represent the model's parameters.

2. **Random Walk (RW) -** We consider the RW(ρ) model from Atkeson et al. [2001]. RW(ρ) is a straightforward but powerful model that predicts the next timestamps by taking the average of the last ρ timestamps, using the formula: $\hat{y}_t = \frac{1}{\rho} \sum_{i=1}^{\rho} x_{t-i} + \varepsilon_t$.

3. **Random Forests (RF) -** The RF(ρ) model is an ensemble learning approach that constructs multiple decision trees [Song and Ying, 2015] to reduce overfitting and enhance generalization [Breiman, 2001]. During prediction, the model returns the average of the predictions made by each individual tree. The inputs to the RF(ρ) model are the last ρ samples, and the output is the predicted value for the next timestamp.

4. **Extreme Gradient Boost (XGBoost) -** The XGBoost(ρ) model [Chen and Guestrin, 2016] is based on an ensemble of decision trees which are trained in a stage-wise fashion similar to other boosting models [Schapire, 1999]. Unlike RF(ρ) which averages the prediction of multiple decision trees, the XGBoost(ρ) trains each tree to minimize the remaining residual error of all previous trees. At prediction time, the

sum of predictions of all the trees is returned. The inputs to the XGBoost(ρ) model are the last ρ samples and the output is the predicted value for the next timestamps.

5. **Fully Connected Neural Network (FC) -** The FC(ρ) model is a fully connected neural network with one hidden layer of size 100 and a ReLU activation function [Ramachandran et al., 2017]. The output layer does not use any activation function to frame the task as a regression problem, optimized using a squared loss function. The inputs to the FC(ρ) model consist of the last ρ samples, and the output is the predicted value for the next timestamp.

6. **Support Vector Regression (SVR) -** SVR(ρ) is a machine learning model based on Support Vector Machines (SVM), used for regression tasks Drucker et al. [1996]. SVR(ρ) attempts to find a function that fits the data within a certain margin of tolerance, while minimizing the prediction error outside this margin. It is particularly effective for capturing complex relationships in data and is robust to outliers due to its focus on maximizing the margin around the prediction. The kernel used for the prediction is "rbf" and the degree of the polynomial kernel function is three. The inputs to the SVR(ρ) model are the last ρ samples and the output is the predicted value for the next timestamps.

5.2 *Ablation Models*

To highlight the impact of the hierarchical component in the Bidirectional HRNN model, we performed an ablation study by comparing it to "simpler" alternatives, specifically GRU-based models that exclude the hierarchical component:

1. **Single GRU (S-GRU) -** The S-GRU(ρ) is a single GRU unit that receives the last ρ values as inputs in order to predict the next value. In GRU(ρ), a single GRU is used for all the time series that comprise the CPI hierarchy. This baseline utilizes all the benefits of a GRU but assumes that the different components of the CPI behave similarly and a single unit is sufficient to model all the nodes.

2. **Independent GRUs (I-GRUs) -** In I-GRUs(ρ), we trained a different GRU(ρ) unit for each CPI node. The S-GRU and I-GRU approaches represent two extremes: The first attempts to model all the CPI nodes with a single model, while the second treats each node separately.

   To emphasize the effect of bidirectionality, we also compared the Bidirectional HRNN to its predecessor, the Hierarchical Recurrent Neural Network (HRNN).

3. **Hierarchical Recurrent Neural Network (HRNN) -** In HRNN(ρ), we trained a separate GRU(ρ) unit for each CPI node, while incorporating the model weights of